

CMAN

Task 1

Given that:

$$S = \max(|W|) / 127$$

$|W| =$

0.85	1.20	0.34	2.10
0.07	0.91	1.88	0.12
1.55	0.03	0.44	2.31
0.18	1.03	0.77	0.55

$$\max(|W|) = 2.31$$

$$S = 2.31 / 127 = 0.018188$$

Task 2 — Quantize

Given that $W_q = \text{round}(W/S)$, clamp to $[-128, 127]$

Row 1:

$$0.85 / 0.018188 = 46.734 \rightarrow 47$$

$$-1.20 / 0.018188 = -65.977 \rightarrow -66$$

$$0.34 / 0.018188 = 18.693 \rightarrow 19$$

$$2.10 / 0.018188 = 115.460 \rightarrow 115$$

Row 2:

$$-0.07 / 0.018188 = -3.848 \rightarrow -4$$

$$0.91 / 0.018188 = 50.032 \rightarrow 50$$

$$-1.88 / 0.018188 = -103.364 \rightarrow -103$$

$$0.12 / 0.018188 = 6.597 \rightarrow 7$$

Row 3:

$$1.55 / 0.018188 = 85.221 \rightarrow 85$$

$$0.03 / 0.018188 = 1.649 \rightarrow 2$$

$$-0.44 / 0.018188 = -24.191 \rightarrow -24$$

$$-2.31 / 0.018188 = -127.006 \rightarrow -127$$

Row 4:

$$-0.18 / 0.018188 = -9.896 \rightarrow -10$$

$$1.03 / 0.018188 = 56.630 \rightarrow 57$$

$$0.77 / 0.018188 = 42.335 \rightarrow 42$$

$$0.55 / 0.018188 = 30.239 \rightarrow 30$$

All values fall within $[-128, 127]$

W_q =

47	-66	19	115
-4	50	-103	7
85	2	-24	-127
-10	57	42	30

Task 3 — Dequantize

$$W_{\text{deq}} = W_q \times S, S = 0.018188$$

Row 1:

$$47 \times 0.018188 = 0.8548$$

$$-66 \times 0.018188 = -1.2004$$

$$19 \times 0.018188 = 0.3455$$

$$115 \times 0.018188 = 2.0916$$

Row 2:

$$-4 \times 0.018188 = -0.0727$$

$$50 \times 0.018188 = 0.9094$$

$$-103 \times 0.018188 = -1.8733$$

$$7 \times 0.018188 = 0.1273$$

Row 3:

$$85 \times 0.018188 = 1.5459$$

$$2 \times 0.018188 = 0.0363$$

$$-24 \times 0.018188 = -0.4365$$

$$-127 \times 0.018188 = -2.3098$$

Row 4:

$$-10 \times 0.018188 = -0.18188$$

$$57 \times 0.018188 = 1.0367$$

$$42 \times 0.018188 = 0.7638$$

$$30 \times 0.018188 = 0.5456$$

W_deq (FP32) =

0.8548	-1.2004	0.3455	2.0916
-0.0727	0.9094	-1.8733	0.1273
1.5459	0.0363	-0.4365	-2.3098
-0.1818	1.0367	0.7638	0.5456

Task 4 — Error Analysis

$|W - W_{\text{deq}}|$

Row 1:

$$|0.85 - 0.8548| = 0.0048$$

$$|-1.20 - (-1.2004)| = 0.0004$$

$$|0.34 - 0.3455| = 0.0055$$

$$|2.10 - 2.0916| = 0.0084$$

Row 2:

$$|-0.07 - (-0.0727)| = 0.0027$$

$$|0.91 - 0.9094| = 0.0006$$

$$|-1.88 - (-1.8733)| = 0.0067$$

$$|0.12 - 0.1273| = 0.0073$$

Row 3:

$$|1.55 - 1.5459| = 0.0041$$

$$|0.03 - 0.0363| = 0.0063$$

$$|-0.44 - (-0.4365)| = 0.0035$$

$$|-2.31 - (-2.3098)| = 0.0002$$

Row 4:

$$|-0.18 - (-0.1818)| = 0.0018$$

$$|1.03 - 1.0367| = 0.0067$$

$$|0.77 - 0.7638| = 0.0062$$

$$|0.55 - 0.5456| = 0.0044$$

Error Matrix =

0.0048	0.0004	0.0055	0.0084
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0.0027	0.0006	0.0067	0.0073
0.0041	0.0063	0.0035	0.0002
0.0018	0.0067	0.0062	0.0044

Largest error = 0.0084 at (Row 1, Col 4)

$W = 2.10$, $W_{\text{deq}} = 2.0916$

Mean Absolute Error (MAE) = Sum of all errors / 16

Sum of all errors = $0.0048 + 0.0004 + 0.0055 + 0.0084 + 0.0027 + 0.0006 + 0.0067 + 0.0073$
 $+ 0.0041 + 0.0063 + 0.0035 + 0.0002 + 0.0018 + 0.0067 + 0.0062 + 0.0044$

Sum = 0.0696

MAE = $0.0696 / 16 = 0.00435$

Task 5 — Bad Scale Experiment

Given $S_{\text{bad}} = 0.01$, clamp to $[-128, 127]$

Row 1:

$0.85 / 0.01 = 85$

$-1.20 / 0.01 = -120$

$0.34 / 0.01 = 34$

$2.10 / 0.01 = 210 \rightarrow \text{clamp } (127)$

Row 2:

$-0.07 / 0.01 = -7$

$0.91 / 0.01 = 91$

$-1.88 / 0.01 = -188 \rightarrow \text{clamp } (-128)$

$0.12 / 0.01 = 12$

Row 3:

$1.55 / 0.01 = 155 \rightarrow \text{clamp } (127)$

$0.03 / 0.01 = 3$

$-0.44 / 0.01 = -44$

$-2.31 / 0.01 = -231 \rightarrow \text{clamp } (-128)$

Row 4:

$-0.18 / 0.01 = -18$

$1.03 / 0.01 = 103$

$0.77 / 0.01 = 77$

$$0.55 / 0.01 = 55$$

W_q_bad =

	85	-120	210 127 ← clamped (was 210)	
	-7	-188 ← clamped (was -188)		12
127 ← clamped (was 155)		3	-231 ← clamped (was -231)	
	-18	103	77	55

Dequantize: W_deq_bad = W_q_bad × 0.01

0.85	-1.20	0.34	1.27
-0.07	0.91	-1.28	0.12
1.27	0.03	-0.44	-1.28
-0.18	1.03	0.77	0.55

Error = |W - W_deq_bad|

0.00	0.00	0.00	0.83
0.00	0.00	0.60	0.00
0.28	0.00	0.00	1.03
0.00	0.00	0.00	0.00

$$\text{Sum of errors} = 0.83 + 0.60 + 0.28 + 1.03 = 2.74$$

$$\text{MAE_bad} = 2.74 / 16 = 0.17125$$

When the scale factor S is too small, values with large magnitudes exceed the representable range and are clamped to the boundary limits of [-128, 127], resulting in irreversible information loss. Consequently, the dequantized values deviate significantly from their original counterparts, producing large quantization errors. This demonstrates that an improperly chosen scale factor severely degrades reconstruction accuracy.